Kurdish Sorani Retrieval: Evaluated System and Future Components

Evaluated raw-text char-4gram BM25 + fine-tuned E5 + zero-shot BGE-M3 fusion; later stages are not implemented

EVALUATED · RAW-TEXT HYBRID RETRIEVAL

INPUT QUERY q — illustrative example

پایتەختی میرنشینی بابان لە چ ساڵێکدا گواستراوەتەوە؟

Gloss: "In which year was the capital of Baban transferred?"

KNOWLEDGE CORPUS C — 492 news passages

Kurdish news articles (headline = query, body = passage)

One article body = one passage · no chunking in our benchmark

Corpus integrity: SHA-256 verified

RAW TEXT — USED IN THE FUSED RETRIEVER

No clitic splitting is applied in the final fused retriever.

The rule-based splitter was tested separately and never selected during validation.

DENSE SCORES — FINE-TUNED E5 + ZERO-SHOT BGE-M3

Query tower

$$e_q = f_\theta(q) \in \mathbb{R}^d$$

Passage tower

$$e_p = g_\varphi(p) \in \mathbb{R}^d$$

$$S_{fused} = w_1 z(\text{char-4gram BM25}) + w_2 z(\text{fine-tuned E5}) + w_3 z(\text{zero-shot BGE-M3})$$

OUTPUT: FUSED PASSAGE RANKING

Per-query z-score weighted fusion of the three evaluated signals

EVALUATED in this paper (retrieval only)

Ranked passages

STAGE 2 · RERANKER — NOT IMPLEMENTED

VERIFICATION POOL — 20 paired passages

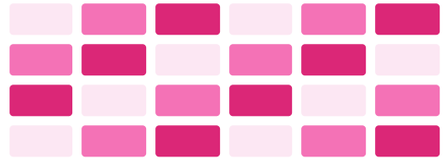
$$\{(q,p_1), (q,p_2), \dots, (q,p_{20})\}$$

Joint token concatenation: [CLS] ∘ q ∘ [SEP] ∘ p

NOT IMPLEMENTED

CROSS-ATTENTION INTERACTION MATRIX

Full all-to-all token-to-token attention weights



$$A_{i,j} = \text{softmax}(QK^T / \sqrt{d_k})$$

⇒ **Deep word-sense disambiguation**

Neutralises lexical surface traps and spurious homonym overlaps

NOT IMPLEMENTED

RANKING & FILTERING — relevance scoring

$$s_{rerank}(q,p) = W_{rerank} \cdot h_{[CLS]} + b$$

High-precision pruning: 20 candidates → top-k = 5 gold context

NOT EVALUATED: no Kurdish reranker was trained or tested

Filters out clitic-inflected semantic distractors

NOT IMPLEMENTED

NOT IMPLEMENTED: NO RERANKED OUTPUT

The box above is a design sketch only.

Future context

STAGE 3 · GENERATOR — NOT IMPLEMENTED

HYPOTHETICAL AUTOREGRESSIVE GENERATOR

Prompt $x = [\mathcal{K}_5(q) \,\|\, q]$ **→ sample N = 5 responses (T = 0.7)**

Stochastic paths: $\mathcal{Y} = \{y^{(1)}, y^{(2)}, y^{(3)}, y^{(4)}, y^{(5)}\}$

Explores the semantic answer distribution to expose epistemic uncertainty

NOT IMPLEMENTED

DUAL-CALIBRATION & FACT-VERIFICATION ENGINE

1 · Semantic entropy (H_{sem})

$$H_{sem} = -\sum_c P_c \log P_c$$

Cluster by NLI entailment: y(i) ⇔ y(j)

2 · Grounding score (Φ_{ground})

$$\Phi_{ground}(y^*, \mathcal{K}_5) \in [0,1]$$

Premise entailment verification

Joint uncertainty decision criterion

Illustrative decision rule; thresholds are untuned

NOT IMPLEMENTED

NOT IMPLEMENTED · EXAMPLE ONLY

Modal consensus answer (y*)

لە ساڵی ١٧٨٤ لەلایەن ئیبراهیم پاشای بابانەوە گواستراوەتەوە

Gloss: "Transferred in 1784 by Ibrahim Pasha of Baban"

Illustrative output (hand-written example, not a measured result)

No generator was run: EM / FactScore not measured

NOT IMPLEMENTED · EXAMPLE ONLY

Calibrated refusal triggered

زانیاری پێویست لە بەردەستدا نییە

Gloss: "I do not have sufficient verified facts in context"

Illustrative refusal text (design only)

Hallucination / calibration: not measured